METHOD STATEMENT

Addmix HPMC — Modified Hydroxypropyl Methyl Cellulose Additive for Dry-Mix Mortars

Project Name		**Document No.**	OMX-MS-ADDMIXHPMC
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	Addmix HPMC	**Status**	For Approval

1. Purpose

This method statement defines the procedure, resources, quality checks and safety measures for the receipt, storage, dosing and quality control of Addmix HPMC in the production of dry-mix mortars.

2. Scope of Work

This method statement covers the receipt, storage, handling, dosing in dry blending and quality control of Addmix HPMC.

- Receipt and inspection of the material.
- Storage and handling in the plant.
- Laboratory trials to set the dosage.
- Dosing and dry blending in production.
- Quality control of finished products.

3. Definitions

Term	Meaning
Dry mix	Factory-blended mortar powder to which water is added on site.
Dosage	Additive quantity as a percentage of the total dry-mix weight.
Water retention	Ability of the fresh mortar to keep its mixing water.
Open time	Time during which tiles or layers can still be placed.

4. Roles & Responsibilities

Role	Responsibility
Production Manager	Overall responsibility for planning and compliance with approved formulations.
Laboratory / R&D	Sets and approves the dosage by trials.
Production Supervisor	Supervises weighing, blending and batch records.
QC Inspector	Tests incoming material and finished products.

Role	Responsibility
HSE Officer	Checks PPE, dust control and safe handling.
Storekeeper	Receives and stores bags under the stated conditions; controls batch numbers.

5. Materials

Material	Description
Addmix HPMC	25 kg bags.

5.1 Key product data

Property	Typical Value
Type	Modified hydroxypropyl methyl cellulose
Appearance	White powder
Bulk density	250 – 550 g/L
Fineness	> 92 % passing 80 mesh
Viscosity (2 % solution)	55,000 – 65,000 mPa·s
Methoxyl content	19 – 24 %
Hydroxypropyl content	4 – 12 %
Shelf life	≥ 24 months

6. Tools & Equipment

- Calibrated scale
- Ribbon or plough-share mixer
- Sieve
- Laboratory mixer
- Flow table
- Test equipment
- Dust extraction at the dosing point
- Sample containers for retained samples

6.1 Personal protective equipment

- Dust mask
- Goggles
- Gloves
- Overalls

7. Storage & Handling

- At least 24 months in original, unopened bags. Store dry, protected from moisture and heat.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Keep containers tightly closed and protect from frost and direct sunlight.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Formulation with approved dosage available.
- Mixer clean and dry.
- Scale calibrated.
- Dust extraction working.
- Team briefed on dosing and dust safety.

9. Laboratory Trial

1. Prepare laboratory batches at three dosage levels within the recommended range.
2. Test water demand, workability, water retention, open time, sag/slip and adhesion as relevant.
3. Select the dosage that meets the product specification.
4. Record the approved dosage in the formulation.

10. Preparation of Material

1. Check the batch number and expiry date and record them.
2. Keep bags closed until use.
3. Weigh the additive on a calibrated scale.

11. Dosing & Dry Blending

1. Charge cement, sand and fillers into the mixer.
2. Add the weighed additive with the other minor components.
3. Blend dry until fully homogeneous (typically 3 – 5 minutes, as per plant procedure).
4. Discharge and pack the dry mix.
5. Record quantities in the batch record.

12. Batch Release

- Test each batch for the key properties.
- Release only when results meet the specification.

- Keep a retained sample of each batch.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Material	On receipt	White powder, dry, within shelf life	QC Lab
Dosage	Each batch	As per formulation	Production Supervisor
Product	Each batch	Within specification	QC Lab

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Appearance, moisture, batch	Each delivery	Goods receipt report
Weighing	Dosage	Each batch	Batch record
Finished product	Workability, water retention, adhesion	Each batch	QC report

14. Defects & Remedies

Defect	Likely Cause	Remedy
Lumps in mortar	Additive added to water; poor blending	Add to dry mix; extend blending
Poor water retention	Under-dosing	Adjust dosage within range
Sticky / heavy mortar	Over-dosing	Reduce dosage

15. Health, Safety & Environment

Hazard	Control Measures
Combustible dust	Avoid dust clouds; no ignition sources; dust extraction
Dust inhalation	Dust masks; extraction
Slippery floors when wet	Clean spills dry first
Manual handling of bags	Correct lifting technique

- Wear a dust mask, goggles and gloves when handling.
- Avoid dust formation.
- Non-hazardous product.
- Eyes: rinse with clean water and seek medical advice if irritation persists.
- Skin: wash with soap and water.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.

- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean equipment with dry methods first, then water.

17. Limitations

- Do not dissolve directly in water.
- Store dry; protect from moisture.
- Any change to the dosage must be approved by the laboratory.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Material received and inspected	☐ Yes ☐ No	
Dosage approved	☐ Yes ☐ No	
Additive weighed	☐ Yes ☐ No	
Dry blended	☐ Yes ☐ No	
Batch tested	☐ Yes ☐ No	
Batch record completed	☐ Yes ☐ No	
Retained sample kept	☐ Yes ☐ No	

Inspected by	Signature	Date